

FlyRank ML Internship 2026

Refresh Opportunity / Content Decay Predictor — Results PDF

30,000 Rows scored Anonymized BigQuery export	random forest Best model Selected by Precision@50	0.747 ROC AUC Ranking separation quality	0.680 Precision@50 Top-50 decline hit rate
3,576 High confidence Rows ready for manual review	0.542 Declining-label rate Supervised target balance	81.9 Top queue score 0–100 final priority score	client holdout Validation style Uses anonymized client split

How to read this quickly

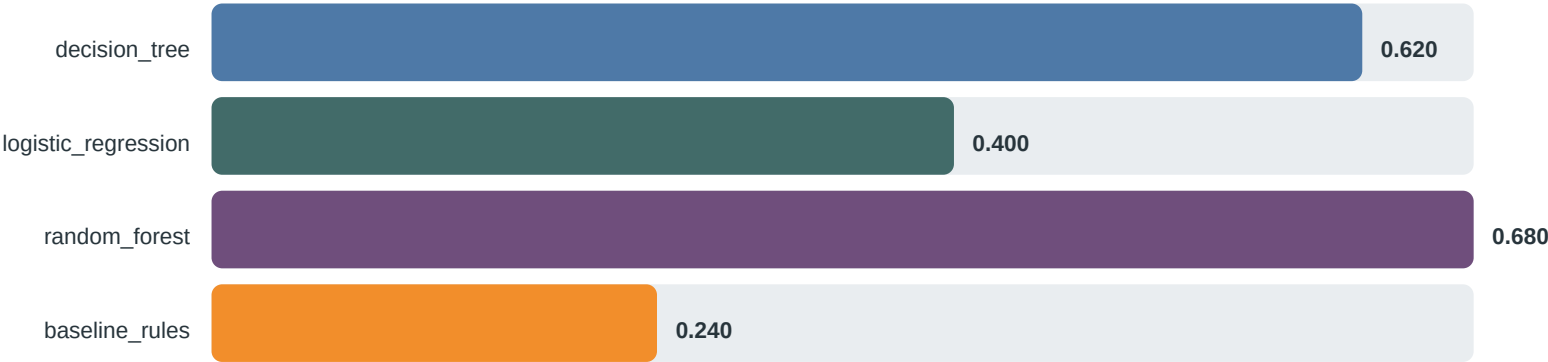
Question	Answer
What is the artifact?	A ranked queue of content rows that should be reviewed for refresh priority.
What is the model doing?	It predicts decline risk, then combines model probability with a non-leaky rule baseline for final prioritization.
What should mentors inspect first?	High-confidence rows with `refresh`, `refresh_and_review_ctr`, or `refresh_and_review_engagement` actions.
What is not included?	No URLs, titles, keywords, domains, client names, or raw private identifiers.

Bottom line: this is a practical decision-support model for the internship capstone. It is not an automatic publishing decision and should be paired with editorial review.

1. Model Quality

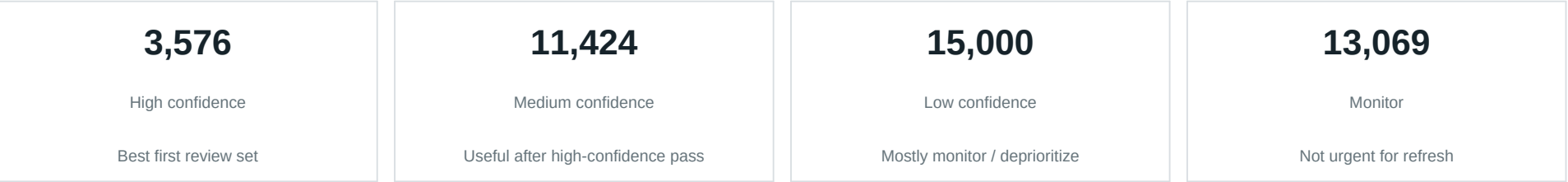
Model	ROC AUC	Avg precision	Precision@50	Recall	F1
decision_tree	0.742	0.575	0.620	0.716	0.634
logistic_regression	0.700	0.522	0.400	0.567	0.566
random_forest	0.747	0.610	0.680	0.741	0.638
baseline_rules	0.627	0.468	0.240	0.189	0.274

Precision@50 comparison

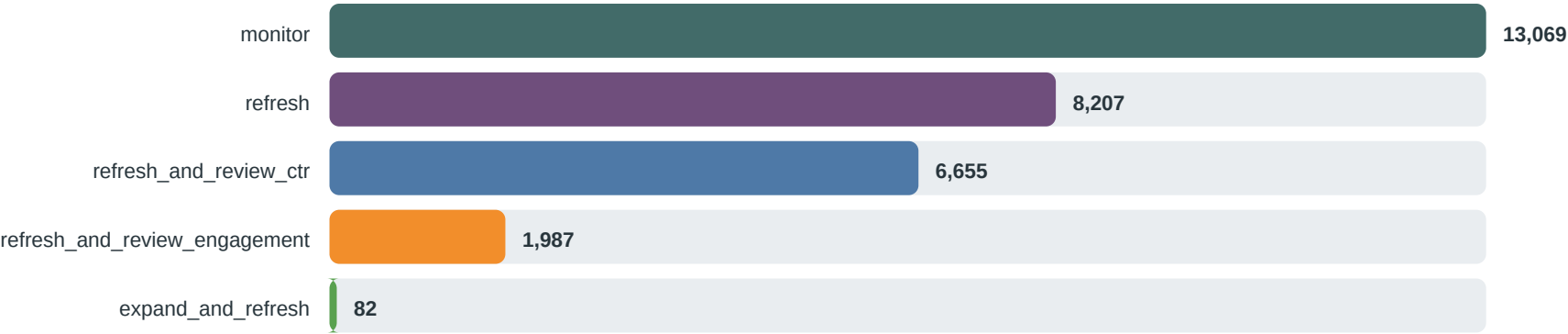


Metric	Plain-English interpretation
ROC AUC	How well the model ranks declining rows ahead of non-declining rows.
Average precision	Quality of the ranked list when positives are mixed with negatives.
Precision@50	How many of the top 50 recommended rows are actually decline-labelled.
Recall	How many decline-labelled rows are captured at the default 0.5 probability threshold.

2. Queue Shape



Suggested action mix

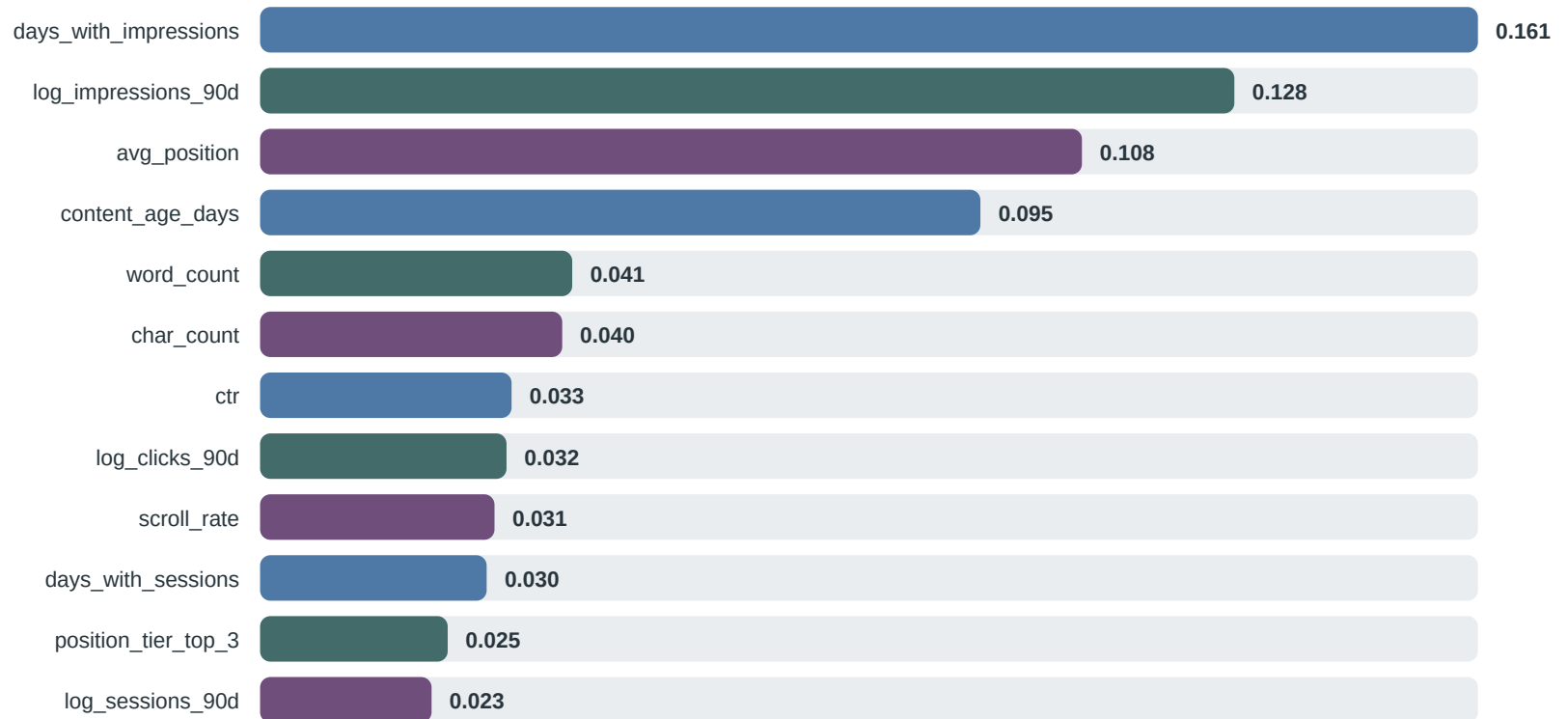


Confidence distribution



3. Why Pages Rank High

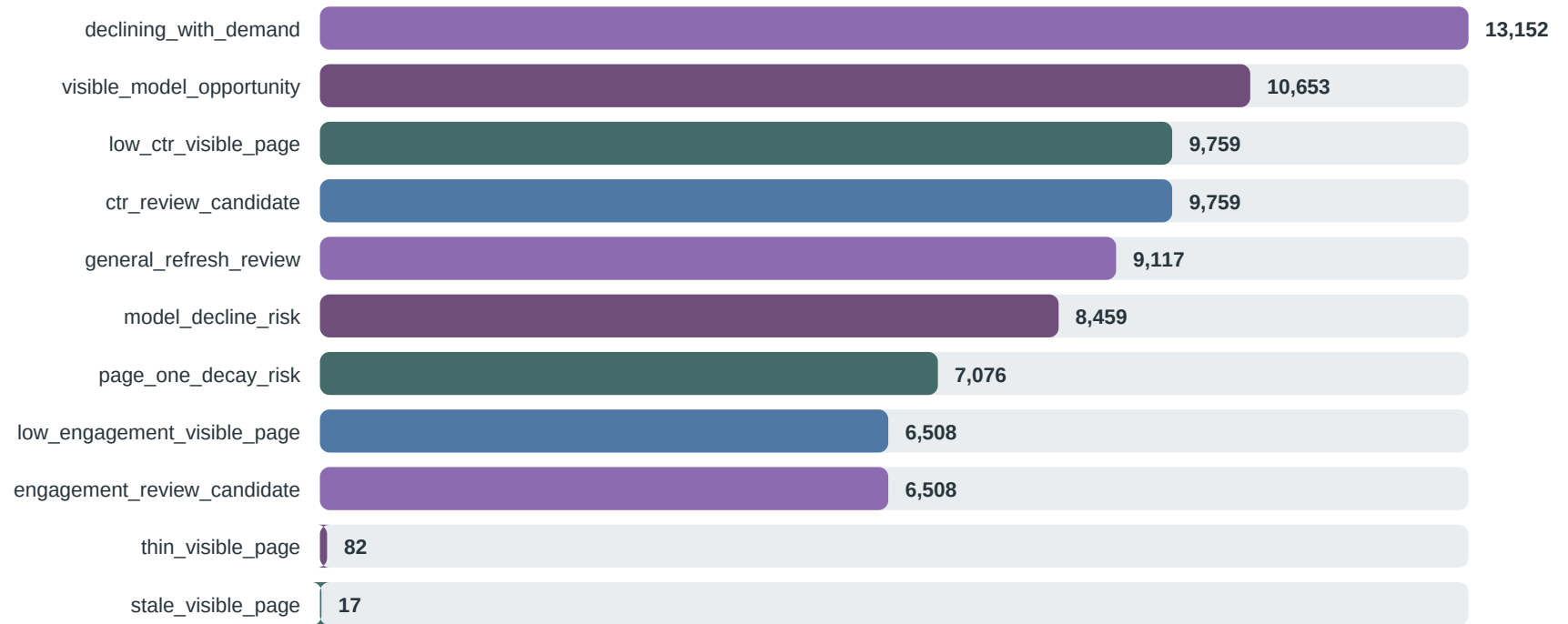
Top model features



Strong signals are mostly visibility and freshness related: days with impressions, recent search visibility, average position, content age, word count, CTR, and engagement-style metrics.

4. Recommendation Reasons

Top recommendation reason codes



Reason codes explain why a row appears in the queue. They are designed for mentor review and intern explainability, not for automatic publishing decisions.

5. Top Queue Preview

These rows are anonymized. Use `content\_id` only to join back internally when mentors are ready to inspect real pages.

Rank	Score	Prob.	Confidence	Action	Impr.	Sess.	Trend	Reasons
1	81.9	0.786	high	refresh + CTR	12,834	66	down	declining with demand, low ctr visible page, low engagement visible page, model decline risk, visible model opportunity, ctr review candidate, engagement review candidate
2	81.7	0.792	high	refresh + CTR	8,064	23	down	declining with demand, low ctr visible page, model decline risk, visible model opportunity, ctr review candidate
3	81.6	0.850	medium	refresh + CTR	2,498	9	down	declining with demand, low ctr visible page, model decline risk, visible model opportunity, ctr review candidate
4	80.8	0.814	medium	refresh + CTR	3,393	5	down	declining with demand, low ctr visible page, model decline risk, visible model opportunity, ctr review candidate
5	80.8	0.771	high	refresh + CTR	13,790	27	down	declining with demand, low ctr visible page, model decline risk, visible model opportunity, ctr review candidate
6	80.8	0.848	high	refresh + CTR	1,622	20	down	declining with demand, low ctr visible page, model decline risk, visible model opportunity, ctr review candidate
7	80.6	0.794	high	refresh + CTR	5,811	14	down	declining with demand, low ctr visible page, model decline risk, visible model opportunity, ctr review candidate
8	80.4	0.828	medium	refresh	4,366	8	down	declining with demand, model decline risk, visible model opportunity
9	80.2	0.824	high	refresh + CTR	2,655	92	down	declining with demand, low ctr visible page, low engagement visible page, model decline risk, visible model opportunity, ctr review candidate, engagement review candidate
10	80.1	0.831	high	refresh + CTR	2,621	10	down	declining with demand, low ctr visible page, model decline risk, visible model opportunity, ctr review candidate

Recommended next validation pass

Step	Validation action
1	Manually inspect the top 25 high-confidence rows in `refresh_queue.csv`.
2	Check whether the recommended action matches editorial context and business priority.
3	If accepted, convert this into the intern capstone task: reproduce, evaluate, explain, and improve the queue.
4	Avoid treating model score as an automatic publishing decision.